

Some initial observations from exploratory data reduction

Wrangling

Some wrangling was necessary before undertaking data reduction. First, I decided to multiply the el* and cw* variables by -1 because they are supposed to measure less-democratic situations. Second, several of our variables are perfectly correlated, so I averaged the perfectly collinear ones (and for the factor analysis, just dropped three more). This left me with c01, c23m, em_c3m, cs_c3m, elcw_c4t5, el_t6t9, cs_t6t9, cw_t6t9, where

$c01 = (el_c0 + cw_c0 + el_c1 + em_c1 + cs_c1) / 5$,

$c23m = (el_c2m + em_c2m + cw_c2m + el_c3m + em_c3m + cs_c3m) / 6$, and

$elcw_c4t5 = (el_c4t5 + cw_c4t5) / 2$.

(Remember, the el* and cw* variables had been made negative.)

Analyses

In the PCA, only the first two components had Eigenvalues > 1, so I restricted the analysis to just those two dimensions. Here are the estimates:

Importance of first k=2 (out of 8) components:

	PC1	PC2
Standard deviation	1.9340	1.4842
Proportion of Variance	0.4675	0.2753
Cumulative Proportion	0.4675	0.7429

Importance of first k=2 (out of 8) components:

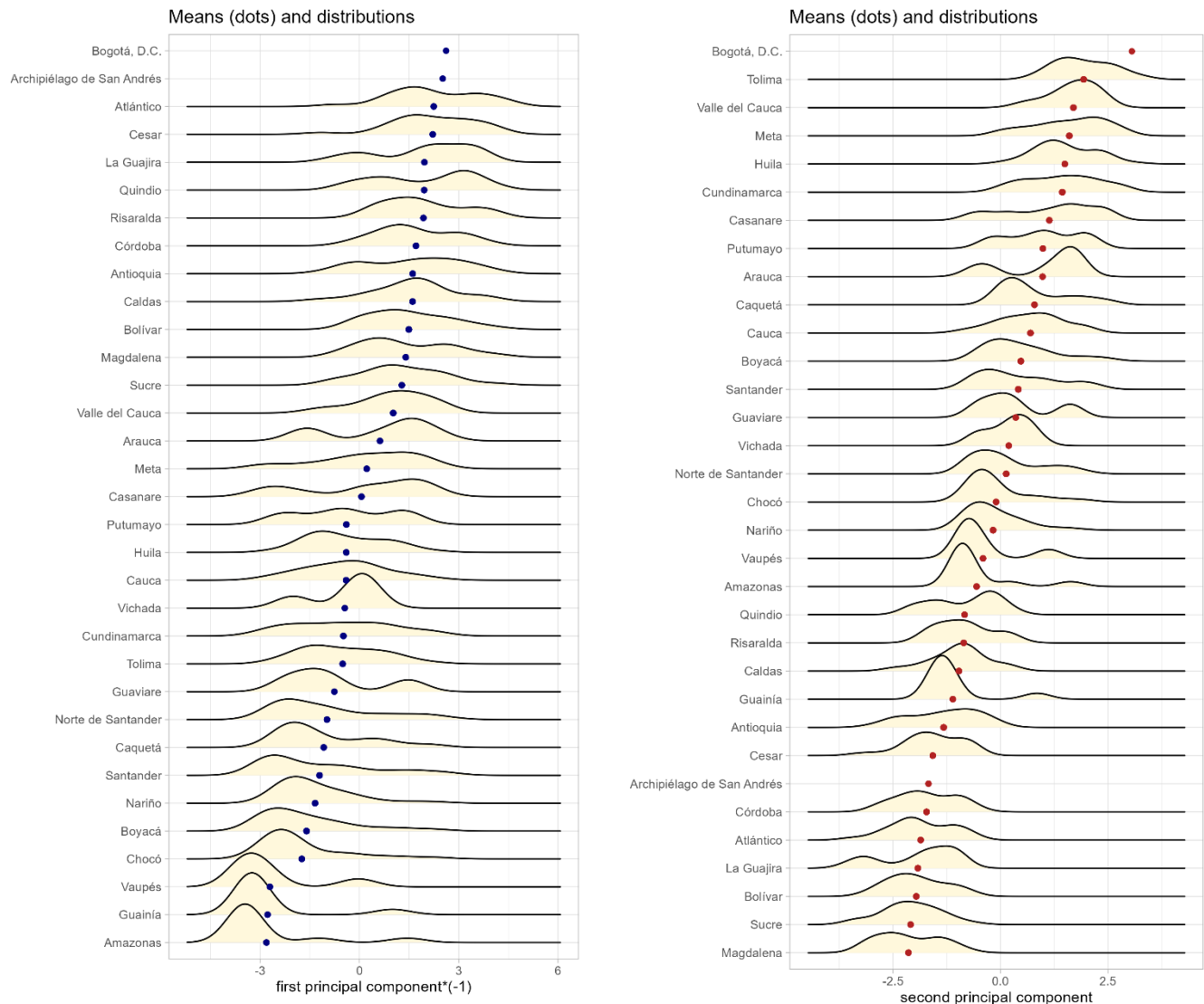
Rotation (n x k) = (8 x 2):

	PC1	PC2
c01	0.3358286	-0.07745605
c23m	0.4640425	-0.27547176
em_c3m	-0.4640425	0.27547176
cs_c3m	-0.4640425	0.27547176
elcw_c4t5	0.0875485	0.40230302
el_t6t9	0.2510118	0.34502194
cs_t6t9	-0.3053098	-0.50579652
cw_t6t9	-0.2780775	-0.47919570

The existence of two significant dimensions, one explaining 47% of the variance and the second 28%, suggests caution about combining these variables with a simple unweighted (= equally weighted) average. Some of the variables suggest a very different ordering of the municipios than others do. Simply averaging all of the variables will do a good job of distinguishing the most-democratic from the least-democratic places, but the intermediate scores will be uninterpretable combinations of high values on some variables and low values on others.

The first component more strongly reflects the rural-urban and low-high income variables, while the second component more strongly reflects ruling party strength and distance from the capital. I predicted scores for each municipio for each component. The signs of the component scores are arbitrary, so I multiplied PC1 by negative 1 to make higher scores correspond to greater democracy. Figures 1 and 2 compare the distributions of the municipal-level component scores by department. There is a lot of variation within departments on the first component, but the extremes (Bogotá vs. Amazonas) correspond to what one would expect based on the rural-urban and low-high income dimensions. On

the second component, the ordering is less obvious, but it seems to reflect proximity to Bogotá and probably the strength or weakness of Duque's party (since this is for 2018).



Figures 1 and 2

However, a scatterplot of the first and second components (Figure 3) subtly suggests a limitation of this PCA. The PCA was based on the assumption that components are orthogonal (uncorrelated) to each other. This shows up Figure 3 as a rotation (evidently in a clockwise direction) that results in a fit line with a slope of zero. (More evidence of this is the fact that the loadings in the table do not clearly distinguish between the first and second components. It would be better if loadings were closer to either 1 or 0.) It would be less restrictive not to assume that the components are orthogonal, which is accomplished by using some kind of oblique rotation. There is a way to do this in a PCA, but I am more comfortable doing it in a factor analysis.

Comparison of first and second principal components

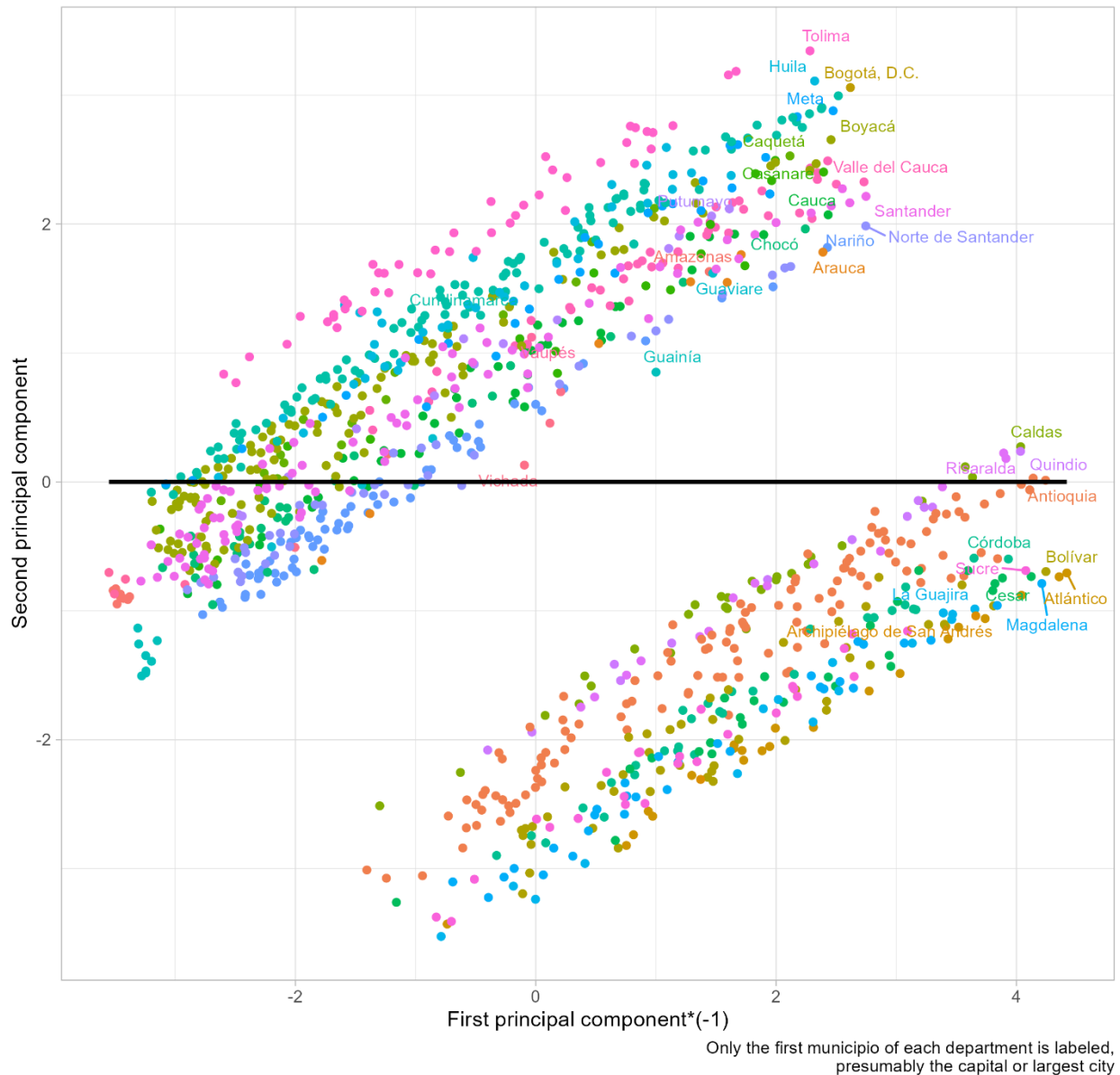


Figure 3

In order to avoid a non-singular matrix in the factor analysis, I had to drop `em_c3m` and `cs_c3m`, which should have been excluded already because they were used in the averaged variable `c23m`. This made it possible to do a factor analysis with a minres model and oblique rotation. There are still two dimensions but they are now correlated at -0.27 and they are switched compared to the PCA: the first factor corresponds best with the second component, and the second factor corresponds best with the first component. The signs are also a little different. Now the second rather than the first factor should be made negative. The estimates are below. The strength or weakness of the ruling party loads very strongly on the first factor, while the rural-urban and low-high income variables load strongly on the second factor. Distance from Bogotá is not as clearly separated between these two dimensions, although it is more related to the first factor. Overall, the factor analysis distinguishes more cleanly

between the two dimensions than the PCA does, although it accounts for only 62% rather than 74% of the variance.

Loadings:

	MR1	MR2
c01	-0.258	0.607
c23m	-0.210	0.861
elcw_c4t5	-0.394	
el_t6t9	-0.538	0.266
cs_t6t9	1.021	-0.281
cw_t6t9	0.892	-0.232

	MR1	MR2
SS loadings	2.395	1.314
Proportion Var	0.399	0.219
Cumulative Var	0.399	0.618

Notice two things about the scatterplot of the two factor scores (Figure 4). First, they are mildly correlated: high scores on the (negative) second dimension correspond slightly with high scores on the first dimension. Second, the first dimension is marked by a clear separation among three distinct levels. This is a result of two conditions. One is that all the three *t6t9 variables (for strength or weakness of the ruling party) have strikingly trimodal or bimodal distributions. The other is that distance from Bogotá has much less influence on this factor than in the PCA. This could be an argument for dropping it altogether, although we should hold off on this until we incorporate other criteria for subnational democracy.

The distributions on these two dimensions are in Figures 5 and 6. They also show strongly ordinal scores on the first factor.

If this were the end of our analysis, we would have three options for dealing with the two-dimensional structure of the data. The easiest one would be to choose one dimension as the “best” or “most valid” one and ignore the other one. If we did this, I would, for now, recommend using the second factor. A more complex choice would be to retain both dimensions and report them separately. I expect this would feel confusing and indecisive to users. The third option would be to find a way to combine the dimensions into an overall index. For this, we would need theoretical guidance to help us decide a) whether they are mutually necessary, which would suggest a multiplicative formula; or substitutive, which would suggest a weighted average; and b) how much additive weight (coefficients in an arithmetic mean) or multiplicative weight (exponents in a geometric mean) each factor score should get. These are hard choices that require careful thought. Faced with this choice in V-Dem, we opted for an average of an additive index and a multiplicative index, but there is no particular reason (yet) to assume that such a hybrid formula would be appropriate for this application.

Comparison of first and second factor score

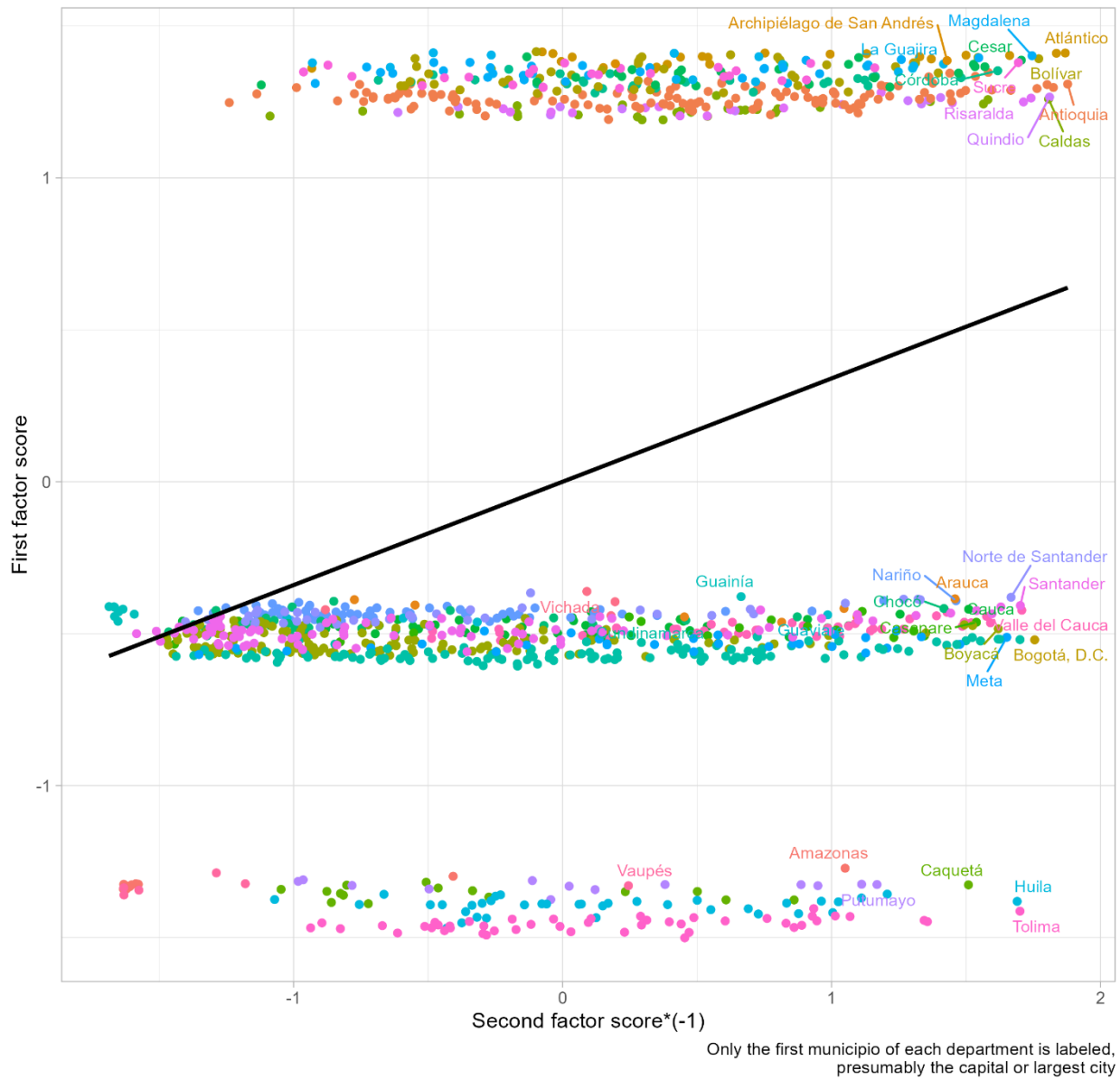
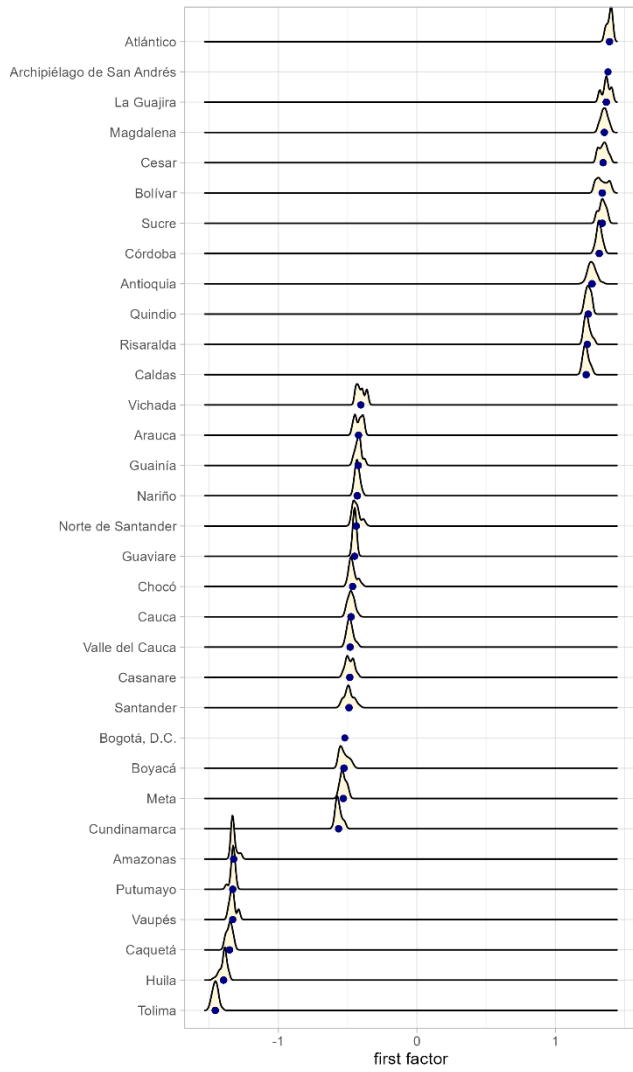
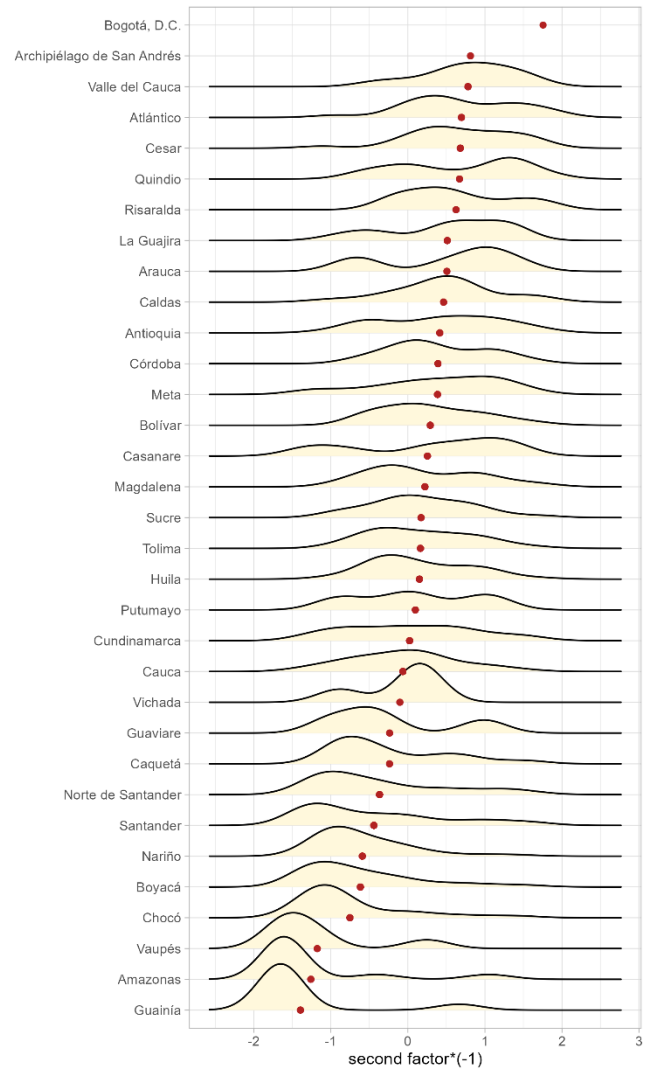


Figure 4

Means (dots) and distributions



Means (dots) and distributions



Figures 5 and 6